# Formula Mass and Percent Composition

Answer key with worked solutions

## A. Understanding

**1.** What percent composition tells you.

> The percentage of the compound's **mass** that comes from each element.

**2.** Why hydrogen is only about 25% of methane's mass.

> Percent composition is by mass, not number of atoms. Hydrogen atoms (1.008 u) are much lighter than carbon (12.01 u): % H = 4(1.008 u) ÷ 16.04 u × 100% = **25.13%**.

## B. Formula mass

**3.** Formula masses.

> (a) $H_2SO_4$: 2(1.008 u) + 32.06 u + 4(16.00 u) = **98.08 u**
>
> (b) $Mg(OH)_2$: 24.31 u + 2(16.00 u + 1.008 u) = **58.33 u**
>
> (c) $(NH_4)_2SO_4$: 2(14.01 u) + 8(1.008 u) + 32.06 u + 4(16.00 u) = **132.14 u**

## C. Percent composition

**4.** Percent composition of NaOH (formula mass 40.00 u).

> % Na = 22.99 u ÷ 40.00 u × 100% = **57.48%**; % O = 16.00 u ÷ 40.00 u × 100% = **40.00%**; % H = 1.008 u ÷ 40.00 u × 100% = **2.52%** (total 100.00%). The units of u cancel in each fraction.

**5.** % N in $NH_4NO_3$.

> Formula mass = 2(14.01 u) + 4(1.008 u) + 3(16.00 u) = 80.05 u. % N = 28.02 u ÷ 80.05 u × 100% = **35.00%**

**6.** Percent composition of ethanol (formula mass 46.07 u).

> % C = 24.02 u ÷ 46.07 u × 100% = **52.14%**; % H = 6.048 u ÷ 46.07 u × 100% = **13.13%**; % O = 16.00 u ÷ 46.07 u × 100% = **34.73%** (total 100.00%)

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

## D. Mass of an element in a sample

**7.** Copper in 25.0 g of $CuSO_4$.

> Formula mass = 63.55 u + 32.06 u + 4(16.00 u) = 159.61 u. Mass of Cu = $\frac{63.55\ \text{u}}{159.61\ \text{u}} \times 25.0\ \text{g} =$ **9.95 g**

**8.** Oxygen in 50.0 g of $CaCO_3$.

> Formula mass = 100.09 u; O contributes 3(16.00 u) = 48.00 u. Mass of O = $\frac{48.00\ \text{u}}{100.09\ \text{u}} \times 50.0\ \text{g} =$ **24.0 g**

## E. Challenge

**9.** Which fertiliser has more nitrogen by mass?

> $NH_3$: 14.01 u ÷ 17.03 u × 100% = **82.25%** N. $(NH_4)_2SO_4$: 28.02 u ÷ 132.14 u × 100% = **21.20%** N. **Ammonia** has the higher mass percent of nitrogen.

**10.** Percentage of water in $CuSO_4 \cdot 5H_2O$.

> Formula mass = 159.61 u + 5(18.016 u) = 249.69 u. % water = 90.08 u ÷ 249.69 u × 100% = **36.08%**

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.